



Capital Budgeting: Sensitivity Analysis

Case

This case was written by Dr. Jasbir Singh Matharu, Associate Professor (Finance) at Institute of Management Technology, Nagpur, India. It is intended to be used as the basis for class discussion rather than to illustrate either effective or ineffective handling of a management situation. The case note was compiled from published sources.

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On a Saturday evening in December 2017, Naini Basu, who was working in the human resources department of a company, was talking to her friend, Priya Solomon, at her house in the Indian city of Nagpur. Naini said, “Priya, I have been working for ten years and accumulated some money. I want to invest in an apartment and rent it out.”

Priya replied, “I am not sure whether this will be a good investment. The rentals are low here; for the past few years, there has been little appreciation in real estate prices (Exhibit 1). But let us find out the net present value and internal rate of return of this investment. I remember studying these under the topic of capital budgeting in my MBA program.”

Naini assembled some information about her proposed investment. She called Priya and said, “The cost of the apartment would be INR 5 million, including stamp duty and taxes. The rules allow depreciation of 5 percent every year on a written down value basis. I expect a rent of INR 180,000 per year. Apartment owners have to pay a property tax of around INR 3000 per year as per the rules of the city’s municipal corporation. Once I buy an apartment in a building, I have to pay a common expense of INR 24,000 for the maintenance of that building. I expect to sell the apartment in ten years for INR 10 million. I currently pay 20 percent tax on my income, and the tax on capital gains is 20 percent.”

Priya explained the capital budgeting process to Naini, “You have to create the profit or loss statement for each year for the next ten years by subtracting the maintenance expenses, property tax, and depreciation from the rent you will receive.”

Naini said, “This is simple; I can do it.” Priya said, “That is not enough. You must add back the depreciation charges to get the cash flows each year because it is a non-cash expenditure.” Priya created the depreciation schedule (Exhibit 2), which shows the yearly depreciation for the next ten years and the reduced value each year after depreciation.

“The next step is to include the cash outflow incurred for buying the apartment. Also, at the end of the tenth year, you will have to include the cash inflow, also called terminal value, on the sale of the apartment. But, remember, if the sale price is more than the book value, you will have to pay tax on capital gains,” Priya explained. For Naini’s understanding, Priya computed the after-tax terminal value (exhibit 3).

Priya explained to Naini, “After computing all the cash flows for ten years, you have to find the net present value (NPV) or the internal rate of return (IRR). NPV is the present value of all cash inflows minus the present value of all cash outflows. The present value calculation is done using the cost of capital. If NPV is positive, the project may be accepted. If it is negative, it means that you will lose money. IRR is the return generated by the investment. If IRR is more than the cost of capital invested, it is worth putting in the money. Here, the cost of capital is the weighted average cost of different sources of capital. Let us assume it to be 10 percent.”

Naini was eager to find out the NPV and IRR of the investment. She put all the data in Excel and found that the NPV was negative (INR -676,333) and the IRR was 8.20 percent which was less than the cost of capital. She felt disheartened. She called up Priya. “Are our assumptions correct or do we assume higher rentals and terminal values?” she asked.

Exhibit 1: Housing Prices in Nagpur city

Quarter	Housing Price for area >646 sq.ft and <=1184 sq.ft (INR)
Dec-13	3,841
Dec-14	4,243
Dec-15	4,264
Dec-16	4,982
Dec-17	5,098

Source: https://residex.nhbonline.org.in/2017-18/NHB_Residex.aspx?cityid=19
(Website of National Housing Bank)

Exhibit 2: Depreciation Schedule

Year	Value, beginning of year, INR	Depreciation every year, INR	Value, end of year, INR
1	5000000	250000	4750000
2	4750000	237500	4512500
3	4512500	225625	4286875
4	4286875	214344	4072531
5	4072531	203627	3868905
6	3868905	193445	3675459
7	3675459	183773	3491686
8	3491686	174584	3317102
9	3317102	165855	3151247
10	3151247	157562	2993685

Exhibit 3: After-Tax Terminal Value

	Amount (INR)
1 Sale Value of the apartment	10,000,000
2 Value as per the books, end of year 10	2,993,685
3 Capital gain on sale (1-2)	7,006,315
4 Tax on capital gain @ 20%	1,401,263
5 After tax resale value (1-4)	8,598,737